Jordan L Vorster

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GitHub: github.com/golba98 | Portfolio: jordanvorsterwebsite.online

Profile

I'm a Computer Science student at the University of London with hands-on experience building developer tools, collaborative web applications and systems software. My projects include a published terminal application, a real-time editor, a Rust compiler and virtual machine, and language models built from scratch in PyTorch and NumPy. My work spans Python, TypeScript, JavaScript, Rust and C++, with practical experience in debugging, automated testing, cloud services and GPU computing.

Education

University of London | *BSc Computer Science, full-time distance learning* *Currently enrolled*

Projects

Ubume CLI | *TypeScript, Bun, Ink* github.com/golba98/Ubume-CLI

- Published an npm package that runs Codex, Claude Code, Antigravity, Mistral Vibe and local models in one terminal UI, with sessions that save, resume and rewind
- Built the human-in-the-loop controls: plan review before anything runs, four permission modes from read-only to full access, and a diff of every file change
- Connected local models (LM Studio, Ollama, Unsloth) to an agent runtime with tool calling, Playwright browser tools and approval before each click or form entry

SyncroEdit | *JavaScript, Cloudflare Workers, Yjs* syncroedit.online

- Built a real-time collaborative editor with Yjs CRDTs over WebSockets, a Durable Object per room, D1 (SQL) persistence and Playwright convergence tests

HP Prime Emulator | *Rust* github.com/golba98/hp-prime-emulator

- Built a compiler and VM for HP Prime programs: Pratt parser, bytecode compiler, stack VM, desktop IDE with a step debugger, and USB testing against the real calculator

HTTP Server | *C++23* github.com/golba98/HTTP-Server

- Wrote an HTTP/1.1 server on POSIX sockets: incremental RFC 9112 parser, sandboxed static files and a bounded thread pool, tested with sanitizers and libFuzzer

IT Pixel Forger | *C++23, Qt 6* github.com/golba98/IT_Pixel_Forger

- Rewrote my Python image-mosaic tool in C++23, making it about 12x faster, with a Qt 6 GUI and animated GIF export

LLM From Scratch (PyTorch) | *Python, PyTorch* github.com/golba98/LLM-From-Scratch

- Built and trained a 900M-parameter decoder-only Transformer in PyTorch across eight packages: BPE tokenizer, data pipeline, training with checkpoint recovery, and inference
- Implemented conversation memory with SQLite storage and cosine-similarity retrieval over embeddings, plus classifier heads on a frozen embedding encoder

Codexa NumPy | *Python, NumPy, CuPy/CUDA* github.com/golba98/Codexa-NumPy

- Built and trained a language model in NumPy with my own tensor/autograd engine, attention, RMSNorm, SwiGLU, cross-entropy and AdamW, plus a CuPy backend for NVIDIA GPUs
- Scaled to 253M parameters with CPU/GPU parity tests, training at 4,000 tokens/sec on an RTX 4080

More on GitHub: 3D Portfolio (TypeScript, React, Three.js), MyGUI (C++20 GUI library on OpenGL) and Forest RPG (procedural world generation in JavaScript)

Work experience

Part-time Intern | *Pam Golding Properties* *2024-2026*

- Photographed homes and prepared listings and documentation for new properties
- Assisted with inventory checks, property value estimates and preparing houses for arrivals
- Worked with agents to troubleshoot issues and streamline daily operations

Summer Crew Member *2021, 2023*

- Prepared rooms and maintained guest and pool areas during peak holiday seasons
- Led a small team to have all guest areas ready on time and to standard

Part-time Waiter *2022*

- Served customers in a busy, high-pressure restaurant

Technical skills

- **Languages:** Python, TypeScript, JavaScript, C++, Rust, C#, Delphi, SQL
- **Web:** HTML, CSS, React, Vue, Angular, Node.js, Express, Bun, Electron, Three.js
- **DevOps & Tools:** Git, GitHub Actions (CI), Docker, Podman, Cloudflare (Workers, D1, Durable Objects, Wrangler), Render, Vercel, Linux (Fedora), CMake, uv, conda, Playwright, Vitest, pytest
- **AI & Data:** PyTorch, NumPy, CuPy/CUDA, Transformer architecture, BPE tokenization, embeddings, cosine-similarity retrieval, SQLite
- **Agents & LLM Apps:** Tool calling, human-in-the-loop approval, permission modes, session state, conversation memory, embedding retrieval, OpenAI-compatible and Anthropic APIs, JSON
- **LLMs & AI Tools:** vLLM, llama.cpp, MLX, Ollama, LM Studio, Unsloth, Hugging Face TRL

Leadership and involvement

Debating Society *Grade 8-12*

Chess Team | *Represented the school in inter-school competitions*

Programming Team | *Co-Leader, mentored peers on team projects*

South African Mathematics Olympiad (SAMO) | *Participant* *Grade 9-11*

Certifications

- **CS50: Introduction to Computer Science** | *Harvard University (edX)*: C, Python and algorithms
- **Introduction to Cloud Computing** | *Coursera*
- **Introduction to Calculus / Calculus Fundamentals** | *Coursera*

Development practices

I use Codex, Claude Code and Gemini to support development, with an AGENTS.md file documenting the HP Prime emulator's crates and build and test commands. Changes must pass Rust tests in CI. I also run differential tests that compare the same program's output on the emulator and my physical HP Prime calculator.

Interests

- Building things, from software tools and games to custom PCs
- Running, benchmarking and fine-tuning local LLMs
- Maths
- The mathematics behind language models, reasoning and intelligent systems